

# RACER-GT Protocol and Preregistration Template

What to Lock Before Any Data Are Seen

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## Abstract

This template is meant to be filled in, timestamped, and deposited *before* collection begins. Its purpose is narrow and specific: to make the retrieval-side researcher degrees of freedom visible. A Google Trends series constructed after the analyst has seen the financial results it will be used to explain is not a measurement; it is a choice. Everything below exists to make that distinction checkable by someone else.

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## 8 Deviations

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# 1 Study identification

- Title, authors, affiliations, contact.
- Deposit location and timestamp (OSF, AEA RCT Registry, institutional repository, or a signed Zenodo deposit).
- RACER-GT version used.
- Protocol hash produced by `racergt design`. Record it here. Any later change to a locked setting changes this hash, which is what makes drift detectable rather than deniable.

# 2 Construct and query specification

State the economic construct first and the query second, in that order, because the reverse order invites choosing whichever query gives the desired result.

- The construct the series is intended to measure.
- Why this keyword or Topic measures it, and what else it plausibly picks up.
- `keyword`, `topic_or_term`, `geo`, `category`, `search_property`, `language`.
- Justification for the category choice. Restricting to Finance changes the construct; so does not restricting. Say which you chose and why.
- Historical start and end dates, and the baseline window for normalization.

# 3 Collection design

- Collection-day offsets, streams, technical replicates, time slots.
- What each stream holds fixed: device, browser profile, cookie state, account state, network route, IP.
- Declaration: an estimated stream effect will not be reported as an IP effect unless a crossover factorial design varying IP alone was run.
- Chunk window length, step, and minimum overlap.
- The collection route (official API, manual export, institution-approved client) and its compliance basis.

## 4 Analysis settings, locked in advance

Every field of the configuration is part of the protocol. State at minimum: calibration threshold  $c_{\min}$ , Huber constant, reference-chunk strategy, normalization mode, duplicate thresholds, covariance estimator, weight cap, G-study transformations, benchmark mode and weights.

## 5 Acceptance thresholds

Fill in every threshold *before* collection, and commit to reporting the resulting status whatever it is.

| Rule                            | What it protects against                                               | Your value |
|---------------------------------|------------------------------------------------------------------------|------------|
| min_unique_pulls                | a batch that is mostly duplicates<br>dependence counted as replication |            |
| min_spectral_effective_pulls    |                                                                        |            |
| max_zero_share                  | daily frequency unsupportable<br>for a rare term                       |            |
| min_level_generalizability      | levels that do not reproduce                                           |            |
| min_innovation_generalizability | daily changes that do not reproduce                                    |            |
| min_detection_kappa             | unstable zero/non-zero pattern<br>one dependence cluster dominating    |            |
| max_component_share             |                                                                        |            |
| max_convergence_mae_100         | consensus not yet converged                                            |            |
| max_benchmark_standardized_rmse | daily and weekly series disagreeing                                    |            |

## 6 Pilot separation

State explicitly whether an initial monitoring window will be used, and commit to the following rule: if monitoring leads to *any* change in thresholds, stitching rules, query, code, or the decision tree, the monitored data become pilot data and the confirmatory batch restarts from a new Day 0. Without this commitment the same data design and validate the acceptance rule, which produces acceptance bias that no amount of subsequent care removes.

## 7 Downstream use

- The financial model the index will enter, specified now.
- Whether it enters in levels or differences. If in differences, the innovation generalizability threshold is the binding one, not the level one.
- How measurement error will be handled: reliability-corrected OLS, SIMEX, or both.
- Declaration: GT processing rules are locked before any return, volatility, spread, or VaR result is examined.

## 8 Deviations

Reserve a section, to be completed after the fact, recording every departure from this protocol, when it was decided, and why. A protocol with an honest deviations section is more credible than one with none.